7a	splitting up of a (large) nucleus / nuclide after absorbing a neutron	B1
bi	(neutrons) one of the products prompt further fission or neutrons collide with further nuclei	B1
bii	The number of neutrons emitted will be very large as the stages proceed and hence leading to a very large number of reactions	B1
biii	92 – 36 = 56	A1
biv	The range of percentage yield varies over a wide range of magnitude and hence in order to show the variation of the values in a single graph, a logarithmic scale is used.	B1
	Percentage yield of having 99 = 6 % Percentage yield of masses equal = 1.5×10^{-2} % Both values to get 1 mark each Therefore, the ratio = $6/(1.5 \times 10^{-2})$ $= 400$	B1 B1
	Gamma photon	B1
	1.00kg / 235 u = 1.00 kg / 235 x (1.66×10^{-27} kg) $=2.56 \times 10^{24}$	C1 A1
	Energy of fission products = $2.5634 \times 10^{24} \times 167 \times 10^{6} \times 1.6 \times 10^{-19}$ Electrical energy = $0.25 \times 2.5634 \times 10^{24} \times 167 \times 10^{6} \times 1.6 \times 10^{-19}$ $= 1.71 \times 10^{13}$ J	C1 C1 A1
	Power = energy / time $= 1.7124 \times 10^{13} / (24 \times 60 \times 60)$ $= 1.98 \times 10^{8}$ W	 C1 A1
	$^{99}_{42}$Mo and its decay products have a very long half life. This means that the storage of $^{99}_{42}$Mo will have to be in a lead container and put away for a very long time. (Max 2 marks for the next 3 points) On the other hand, $^{140}_{54}$Xe has a very short half life and many decay products. For the same amount of these fission products, $^{140}_{54}$Xe having many more fission products imply a greater safety concern for 2 reasons: 1. the total activity of $^{140}_{54}$Xe and its daughter products is much higher than $^{99}_{42}$Mo . 2. with a short half-life, it means the decay constant is much greater, using $A=\lambda N$, the activity from $^{140}_{54}$Xe is much higher.	B1 B1 B1

	The much higher activity implies that $^{140}_{54}Xe$ is very hazardous immediately after it is produced, and must be handled with great care.	